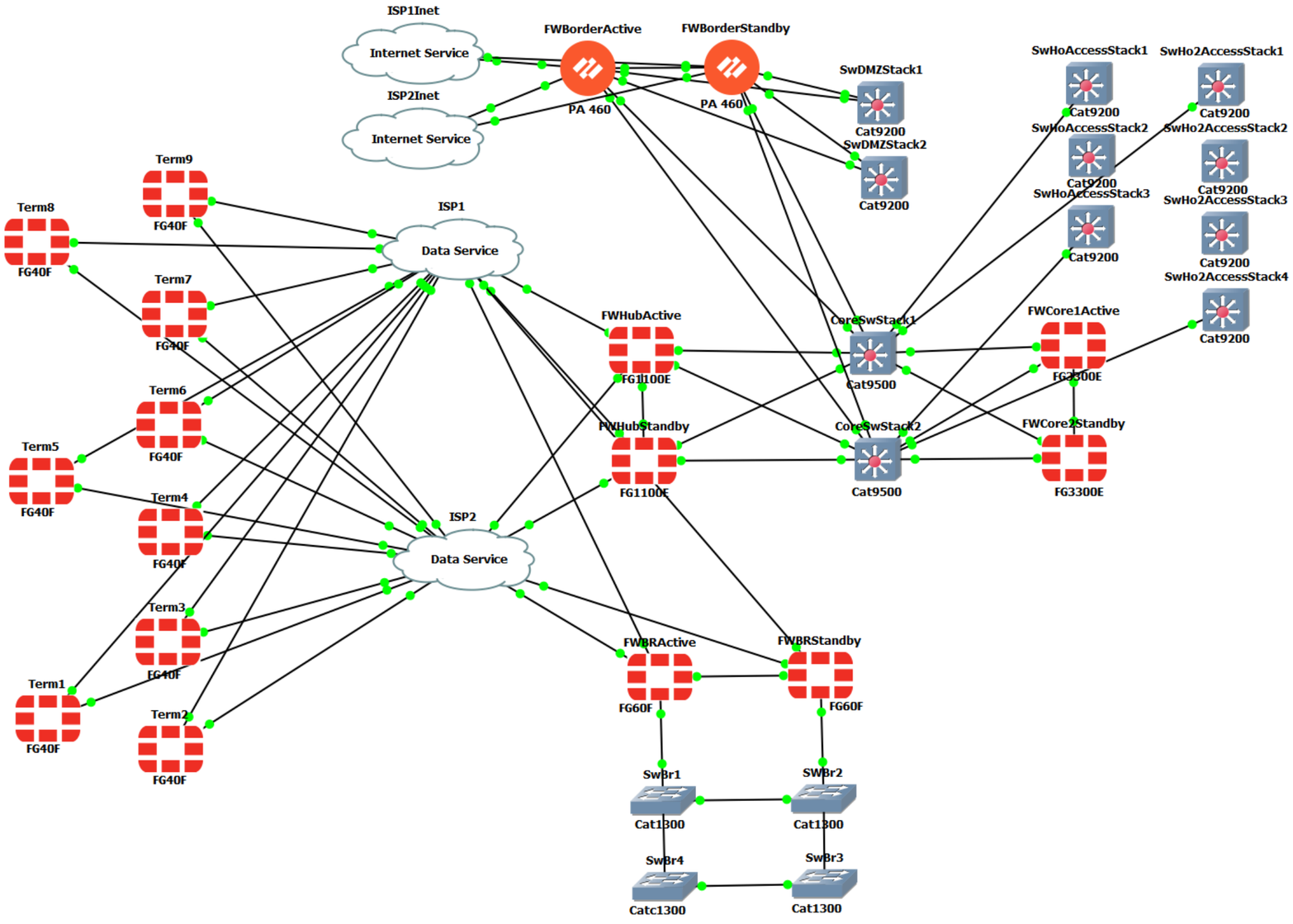
Fintech enterprise network topology



Design legend

Perimeter and DMZ

Two PA-460 firewalls operate in active-passive HA. Each connects to both Internet providers, the Catalyst 9200 DMZ stack and the Catalyst 9500 core. A dedicated transit VLAN connects the PA-460 tier toward the FortiGate 3300E tier. DMZ VLAN gateways reside on the PA-460s.

BGP multihoming

The enterprise advertises its own public prefixes and ASN to both ISPs. ISP1 is preferred inbound by AS-path prepending toward ISP2; outbound default-route preference uses higher BGP Local Preference for ISP1. ISP2 is backup.

Core and internal segmentation

Two Catalyst 9500 switches form a StackWise Virtual core. The core is Layer 2 for protected networks. FortiGate 3300E active-passive HA provides head-office and backend VLAN gateways, security zones and inter-zone enforcement.

VPN hub and remote terminals

FortiGate 1100E active-passive HA terminates dual route-based IPsec tunnels for each remote FortiGate 40F terminal. OSPF operates over both tunnels; higher cost on the backup tunnel favors ISP1.

Branch office access

FortiGate 60F active-passive HA connects to independent Catalyst 1300 switches. Rapid per-VLAN spanning tree provides branch Layer 2 resilience; Switch 1 is preferred root, with Switch 2 taking over on failure.

High availability

Firewall pairs have dedicated HA links, session pickup and monitored LAN interfaces; LACP aggregate interfaces span both Catalyst 9500 core members where shown.

Architecture and implemented behavior reported by project owner | PA-460 BGP capability: Palo Alto Networks PAN-OS 10.2 BGP Import/Export documentation.